



Enhancing Angular Sensitivity of Segmented Antineutrino Detectors

for Reactor Monitoring

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Localization of IBD Source

- Directional method for segmented Inverse Beta Decay (IBD) detectors
- Positron (e^+) gets most of the energy, and annihilation marks prompt vertex
- Neutron (1n) gets momentum direction, and capture marks delayed vertex
- Lithium 6 dopant localizes 1n capture with ^3H and ^4He depositing energy within $\sim 10 \mu\text{m}$ of capture
- direction to source inferred from 1n direction with enough events

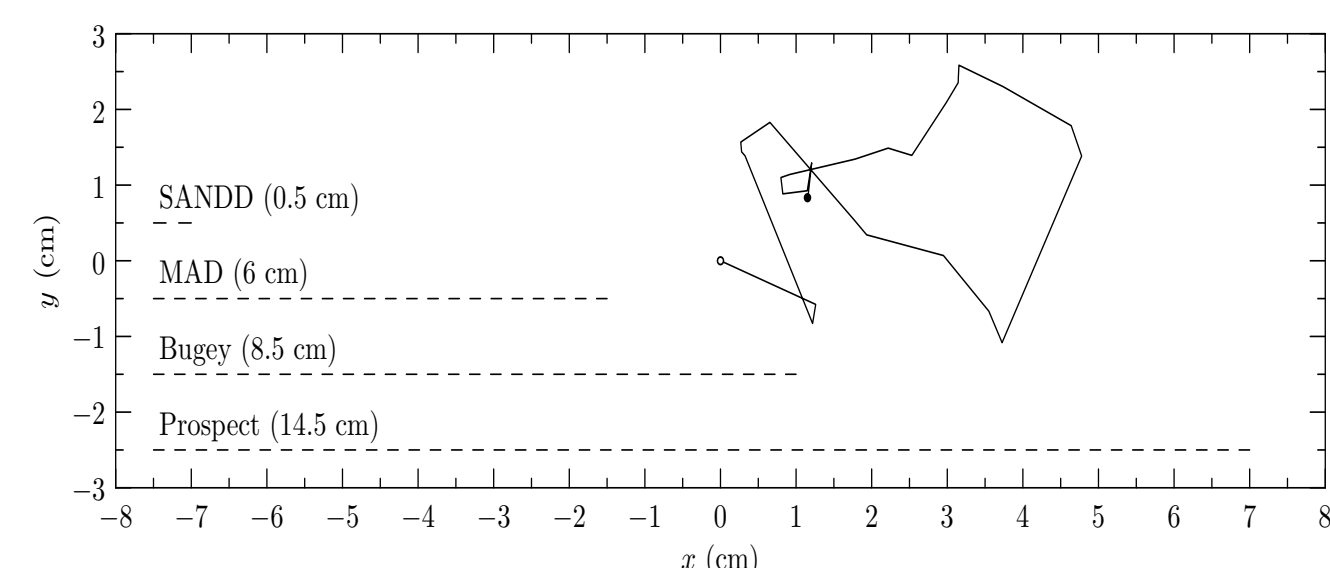


Figure 1: Characteristic neutron xy -trajectory from IBD vertex to capture, drawn to scale against the segment pitch of four 2D-segmented detectors: SANDD, MAD, Bugey, and PROSPECT.

Traditional Approach

- Conventional uncertainty calculation:

$$\Delta\varphi = \arctan\left(\frac{P/l}{\sqrt{N}}\right) \quad (1)$$

- If mean position resolution $P \rightarrow 0$, angular uncertainty $\Delta\varphi \rightarrow 0$ – unphysical!
- Vogel and Beacom model shows a large spread ($\sim 26^\circ$) in initial neutron direction
- Neutrons scatter many times before capture

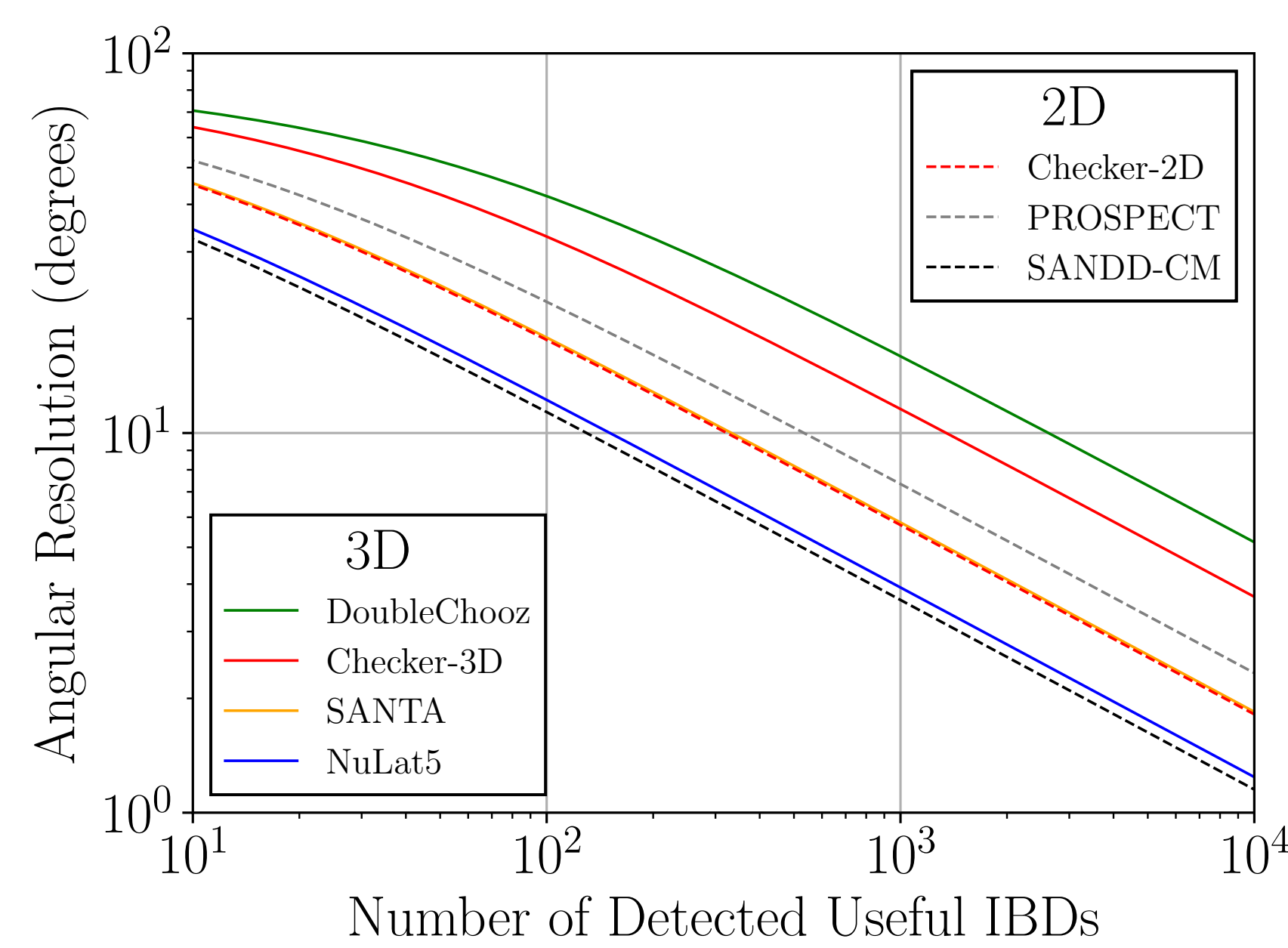


Figure 2: Conventional “Chooz”-style angular uncertainty for a selection of segmented detectors (Chooz highlighted). Useful IBD events have prompt and delayed signals in separate segments. From our previous study [2].

Our Approach to Simulation Design

- Used RATPAC-2 for all simulations
- Created simple monolithic geometry using 0.1%-wt ^6Li -doped plastic scintillator
- Created grid overlay to create binning matrices for 5 mm, 50 mm and 150 mm segment sizes in analysis

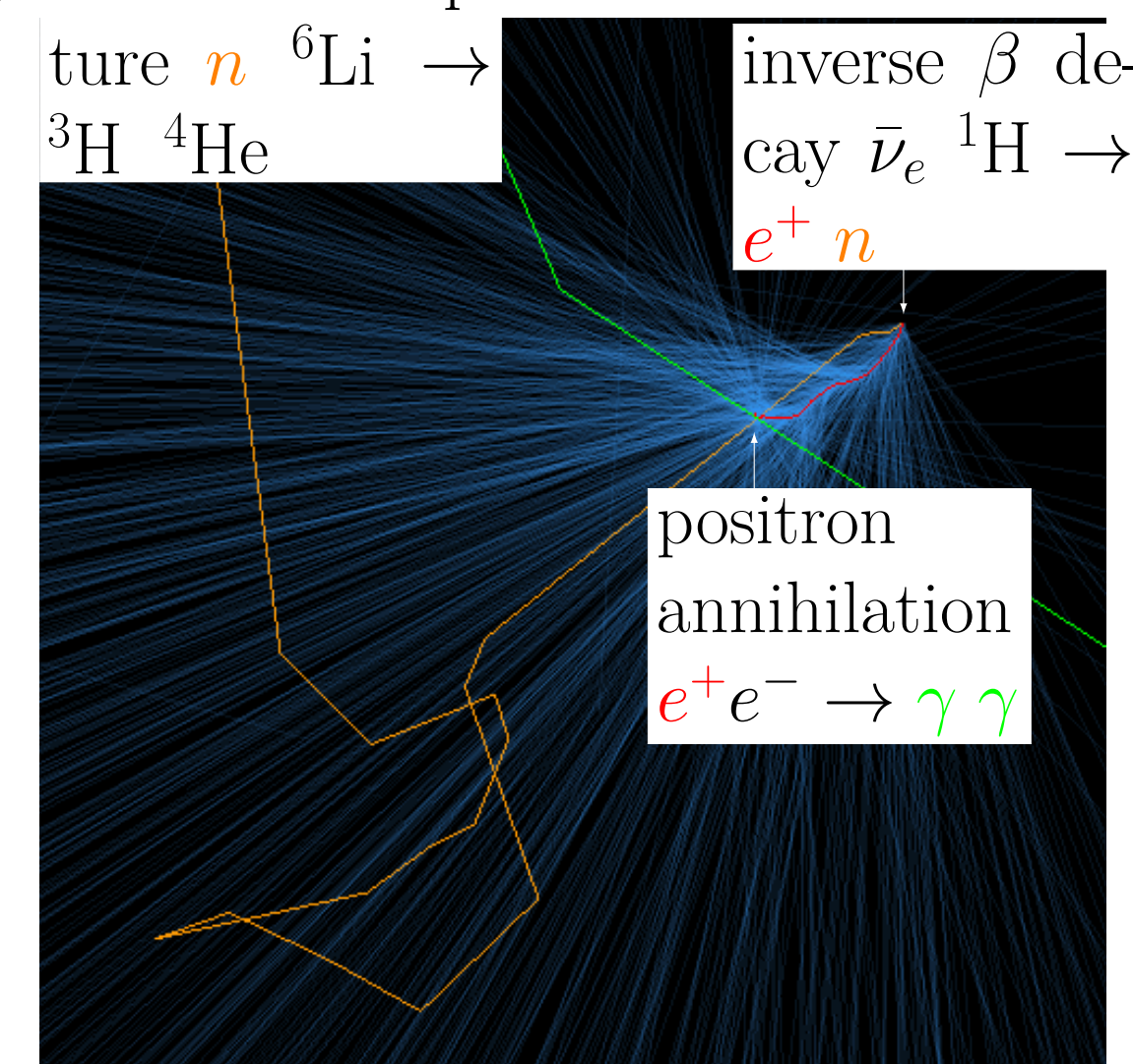


Figure 3: RAT-PAC 2 visualization of one IBD event, scintillation light suppressed for clarity. Tracks: positron (red), neutron (orange), γ -rays (green), Cherenkov photons (cyan). The Cherenkov cone marks the positron direction; here the neutrino and positron travel from top-right to bottom-left.

Novel Algorithm for Segmented Detectors

- Generate reference data for all incident angles
This is done by rotating the grid overlay on simulation data and then performing the binning described below
- Bin prompt and delayed displacement vectors in matrix form by counting the number of voxels separating the IBD vertex and the capture vertex. Normalize all matrices for reference angles
- Create binning matrix for data set and normalize it
- Compute the L^2 norm for the matrix differences for all reference angles
- Fit $\alpha|\sin(\frac{\theta-\theta_0}{2})|$ to L^2 norms as a function of angle. The minimum gives the θ_0 fit parameter for the direction of the neutrino source.

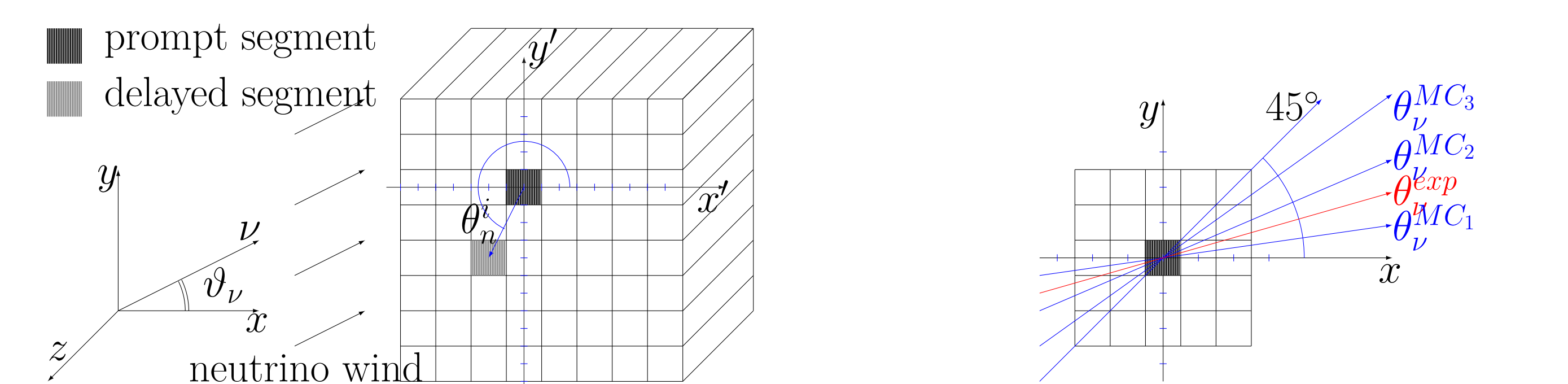


Figure 4: Diagram explaining the prompt-delayed segment geometry, as well as neutrino and axes orientation.

★ Conclusions, Findings, and Future Work

Key findings:

- At large n , the algorithm gives the direction of antineutrino sources
- At small n , the algorithm gives a reasonable uncertainty
- As a pattern matching method, the algorithm provides a comparison between distributions

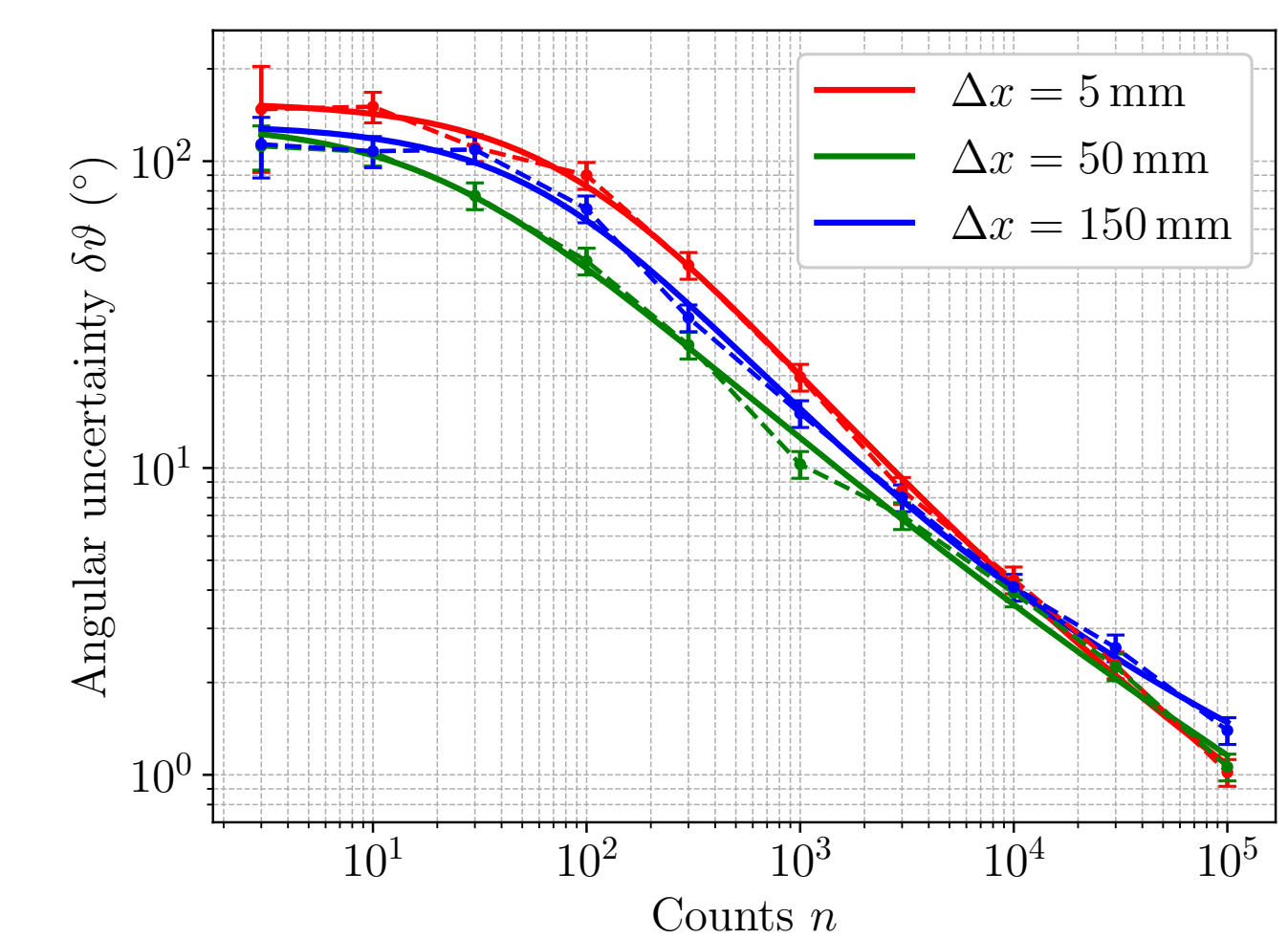
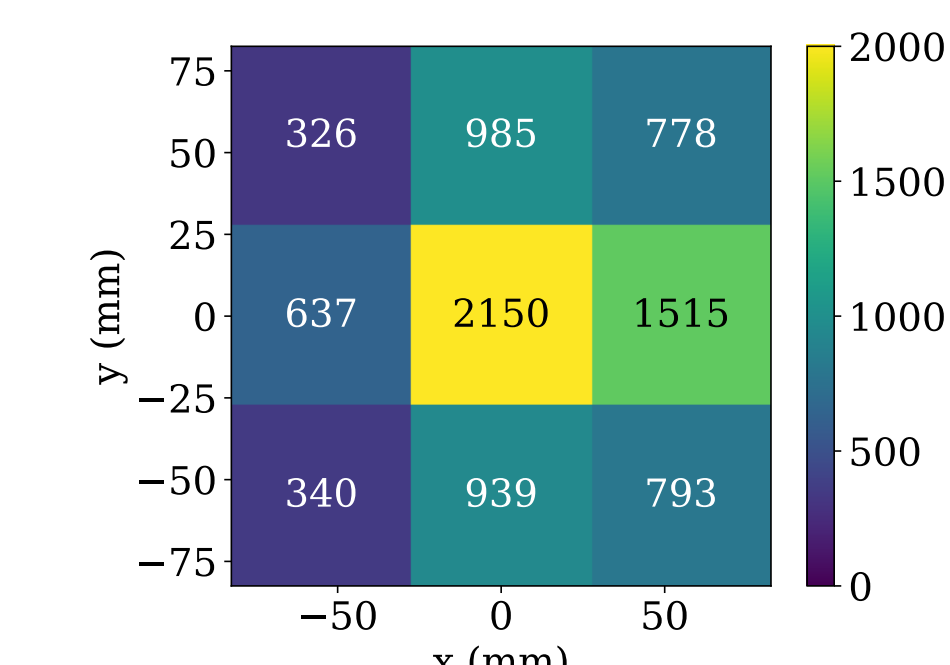
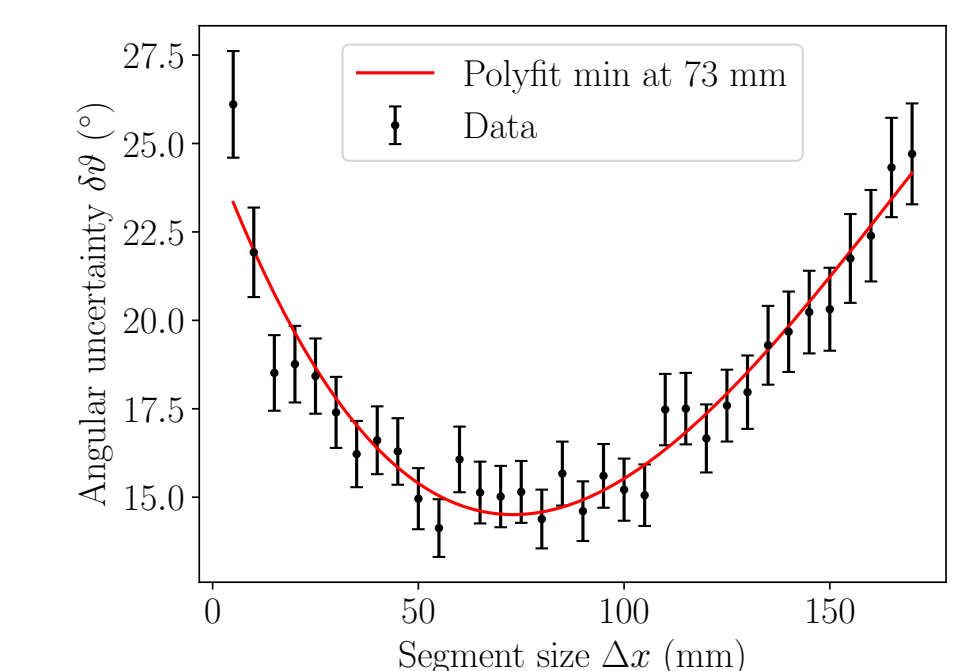


Figure 5: Angular uncertainty $\delta\vartheta$ versus event count n for square segment sizes of 5, 50, and 150 mm.



(a) binning distribution, $\vartheta_\nu = 0^\circ$



(b) angular uncertainty vs. segment size

Figure 6: (a) Binning distribution at $\vartheta_\nu = 0^\circ$ (0.1 wt% ^6Li ; arrow marks the antineutrino direction); other incoming angles rotate the pattern. (b) Angular uncertainty versus segment size Δx , with a polynomial fit minimizing near $\Delta x \approx 73$ mm.

Optimal segment size: $\Delta x \approx 73$ mm

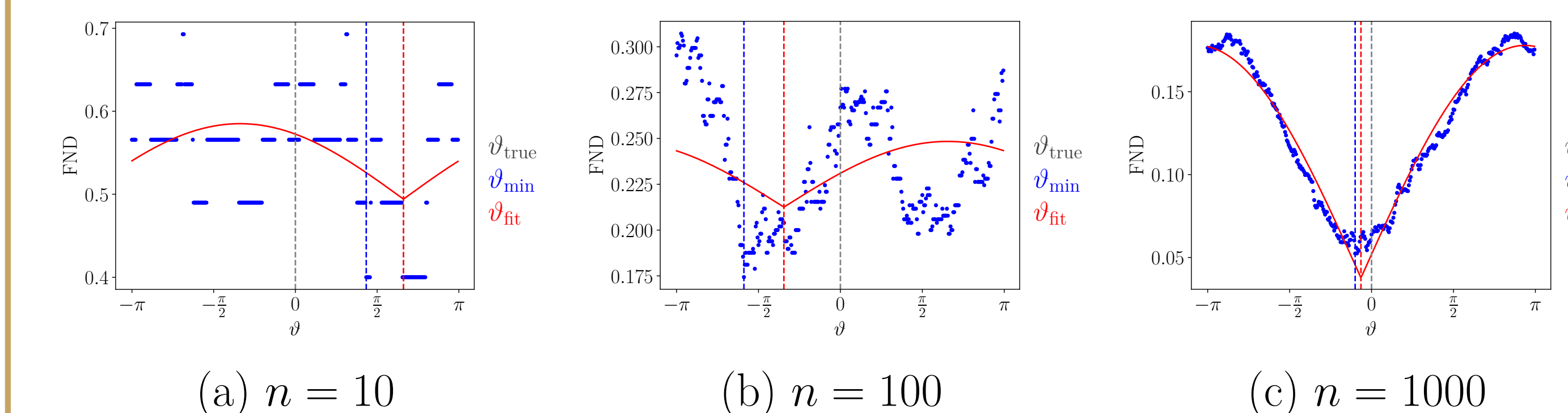


Figure 7: FND versus reference angle ϑ at $n = 10, 100, 1000$ (50 mm segments, RAT-PAC 2 capture data). As n grows, the minimum $\vartheta_{\min}$ and arctan fit ϑ_{fit} approach the true angle ϑ_{true} .

Future work:

- Developing methods to take advantage of machine learning
- Investigating how to apply to monolithic detectors
- Investigating a generalized approach to three dimensionally segmented detectors

References

- [1] Crow *et al.*, in prep. (2026)
- [2] Duvall *et al.*, Phys. Rev. Applied **22**, 054030 (2024)

Acknowledgments

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Ref. [1]:

